

Wenhao Zhang

PhD candidate at National University of Singapore

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Research Interests Machine learning; Human-building interaction; Reinforcement learning; Energy efficient control; Indoor thermal comfort

Education **National University of Singapore** Singapore
PhD in Built Environment Jan 2024 – Present
Supervisors: Prof Clayton Miller, Prof Adrian Chong, Prof Stefano Schiavon.

University College London London, UK
MSc Smart Buildings and Digital Engineering Sep 2022 – Sep 2023
Supervisor: Dr Rui Tang. *Degree: Distinction (Rank: 1/44).*

University of Nottingham China & UK
BEng Hons Architectural Environment Engineering Sep 2017 – Jul 2021
Supervisor: Dr Zhiang Zhang. *Degree: First-Class Honours (Rank: 3/67).*

Honors and Scholarships Gold Medal (Global AI Challenge for Building E&M Facilities) 2025
Most Efficient AI Algorithm Award (Global AI Challenge for Building E&M Facilities) 2025
NUS Research Scholarship (National University of Singapore) 2024
UCL Best Overall Degree Mark Award (University College London) 2023
DesignBuilder Award (DesignBuilder Software Ltd.) 2023
Dean's Scholarship (University of Nottingham) 2021
Head's Scholarship (University of Nottingham) 2020

Journal Publications **Field evaluation of dynamic bedroom air-temperature control for sleep in real homes: implementation fidelity and exploratory sleep outcomes**
Wenhao Zhang*, Adrian Chong, Hui Zhang, Stefano Schiavon, and Clayton Miller.
Building and Environment, 2026. ([Link](#))

Recommender Systems and Reinforcement Learning for Building Control and Occupant Interaction: A Text-Mining Driven Review of Scientific Literature
Wenhao Zhang, Matias Quintana, and Clayton Miller*.
Energy and Buildings, 2024. ([Link](#))

Conference
Publications

Reinforcement Learning in Building Controls: a Comparative Study of Algorithms Considering Model Availability and Policy Representation

Ziyan Wu, **Wenhao Zhang**, Rui Tang*, Huilong Wang, and Ivan Korolija.
Journal of Building Engineering, 2024. ([Link](#))

Scalable Adaptive AC Control in Real Sleep Environments: A Pilot Deployment with Lightweight Reinforcement Learning

Wenhao Zhang*, Adrain Chong, Stefano Schiavon, and Clayton Miller.
ACM BuildSys'26, 2026. ([Link](#))

Towards Uncovering Indoor Satisfaction Profiles: LLM Capacity for Structured Extraction from Occupant Feedback (Best Paper Award)

Tom Parkinson, **Wenhao Zhang**, Stefano Schiavon, and Clayton Miller*.
ACM BuildSys'26, 2026. ([Link](#))

PhD Forum Abstract: The Effects of the Thermal Environment on Sleep and Novel Control Solutions

Wenhao Zhang.
ACM Sustainability Week'26, 2026. ([Link](#))

Bedroom Nighttime Temperature, Humidity and Sleep Duration in Singapore Homes

Michèle Renard, Clayton Miller, Mario Frei, Yun Xuan Chua, Pearl MS Tan, Gabriel May Moe Kya, **Wenhao Zhang**, Toby Jia Xu Seah, Xiaojun Fan, Hui Zhang, Tom Parkinson, June Chi-Yan Lo, Jason Kai Wei Lee*, and Stefano Schiavon*.
Indoor Air 2026, 2026. ([Link](#))

Just-in-Time Adaptive Interventions (JITAI) to Improve Indoor Air Quality in Sleep Environments: A Pilot Study

Wenhao Zhang, Mario Frei, Yun Xuan Chua, Michèle Renard, Xiaojun Fan, Adrain Chong, Tom Parkinson, Jason Kai Wei Lee, Stefano Schiavon, and Clayton Miller*.
Indoor Air 2026, 2026. ([Link](#))

Poster Abstract: Field evaluation of body thermoregulation-based dynamic bedroom air temperature control to improve sleep

Wenhao Zhang, Stefano Schiavon, and Clayton Miller*.
ACM BuildSys'25, 2025. ([Link](#))

Wrist to Rest: A pilot study to characterize sleep habits and bedroom environments using scalable watch-based microsurveys

Clayton Miller*, Yun Xuan Chua, Mario Frei, **Wenhao Zhang**, Gabriel May Moe Kyaw, Toby Jia Xu Seah, Xiaojun Fan, Tom Parkinson, Hui Zhang, June Chi-Yan Lo, Jason Kai Wei Lee, and Stefano Schiavon.
CISBAT 2025, 2025. ([Link](#))

Bedroom ventilation and air quality during sleep: Insights from a pilot field study in Singapore

Xiaojun Fan, **Wenhao Zhang**, Raagavi Mani, Clayton Miller, Tom Parkinson, June Chi-Yan Lo, Jason Kai Wei Lee, Hui Zhang, and Stefano Schiavon*.
Healthy Building Conference 2025, 2025. ([Link](#))

Energy Efficient Operation Optimization of Building Air-conditioners via Simulator-assisted Asynchronous Reinforcement Learning

Wenhao Zhang, and Zhiang Zhang*.
IOP Conference Series Earth and Environmental Science, 2022. ([Link](#))

**Research
Projects**

HEATS: Heat Exposure, AcTivity, and Sleep Field Study

PIs: Prof Stefano Schiavon (UC Berkeley) Jan 2024 – Present
Prof Jason Kai Wei Lee (National University of Singapore)
Prof Clayton Miller (Singapore Management University)
Dr Thomas Parkinson (The University of Sydney)
Prof June Chi-Yan Lo (National University of Singapore)
Prof Hui Zhang (UC Berkeley)

This field study investigates how cumulative heat exposure affects sleep and physical activity among working-age Singaporeans. In this research, a body thermoregulation-based bedroom air temperature control framework and a Just-in-Time Adaptive Intervention (JITAI) model will be developed and evaluated to enhance occupants' sleep in typical Singapore residential buildings.

Comparative Study of Model-Based and Model-Free Reinforcement Learning Control Performance in HVAC Systems

PI: Dr Rui Tang (University College London) May 2023 – Jan 2024
This research conducts the comparison of model-based and model-free reinforcement learning control strategies in HVAC systems. It focuses on analyzing their performance in terms of energy efficiency and indoor comfort. Results indicate both strategies significantly outperform traditional controls, with model-free showing greater resilience to disturbances.

Deep Reinforcement Learning for Energy-Efficient Control of Variable Refrigerant Flow HVAC Systems

PI: Dr Zhiang Zhang (University of Nottingham) Sep 2019 – Jun 2020
Developed an energy efficient operation strategy for VRF system during the cooling season for a case office and achieved energy savings of up to 16.1% as well as improved thermal comfort compared to a rule-based control strategy.

Employment	DesignBuilder Software Ltd.	London, UK
	Technical Writer (Contrator)	Aug 2023 – Jan 2024
	Develop DesignBuilder Scripting Basics training content covering EMS and Python Scripting for runtime, pre and postprocessing of simulations along with DesignBuilder API.	
	China Academy of Building Research	Shanghai, China
	Energy Consultant (Full-time)	Oct 2021 – Aug 2022
	Participated in developing building energy simulation software (PKPM), and the design of ultra-low energy buildings with information technology.	
Services	Web Chair	
	13th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys 2026)	
Skills	Programming	
	Proficient in: Python (scikit-learn, Gym, TensorFlow, Word2Vec, NLTK).	
	Familiar with: AWS Lambda, InfluxDB, C#, MATLAB, Modelica.	
	Languages	
	Chinese (Native), English (Proficient).	